

MUDFISH PROJECT

Matrix Lofting Engine — Current Design Specification

September 15, 2026 | Living Document — append, never overwrite

§1 Purpose and Governing Philosophy

This document is the single source of truth for the Mudfish Matrix Lofting Engine. It captures every design decision made to date and serves as the contract that all future work — spreadsheet additions, matrix cards, formula cards, continuity tests — must honor without exception.

Two rules govern this document:

- Append, never overwrite. New decisions are added as dated sections below existing ones. Nothing already locked is changed without an explicit, dated amendment.
- The uploaded workbook is the reference, not any AI's recollection of it. When in doubt, read the file.

§2 Locked Coordinate Convention

Status: FROZEN. No change permitted without explicit amendment.

+X: Forward (bow direction, positive)

-X: Aft (stern direction, negative)

+Y: Port (left side looking forward) — right-hand rule requires this

-Y: Starboard

+Z: Up (toward sky)

-Z: Down (toward keel)

X = 0: Midship — the longitudinal reference

Z = Z_DWL: Design Waterline — the vertical reference (Z_DWL = 0 in local design frame)

Row-vector kernel contract: $P' = P \times M$. Translation lives in the LAST ROW of every 4×4 matrix. This is the transpose of the standard graphics-textbook column-vector convention.

§3 The Four-Quadrant Mantra

The hull is divided by two fundamental references — Midship (X=0) and the Design Waterline (Z=Z_DWL) — into four geometric quadrants. Every formula card and every matrix card is assigned to exactly one quadrant. No formula crosses a quadrant boundary.

F-U: Forward-Upper (X > 0, Z > Z_DWL) — topsides forward of midship

A-U: Aft-Upper (X < 0, Z > Z_DWL) — topsides aft of midship

F-L: Forward-Lower (X > 0, Z < Z_DWL) — underbody forward of midship

A-L: Aft-Lower (X < 0, Z < Z_DWL) — underbody aft of midship

The DWL boundary is simultaneously the bottom of the Upper cards and the top of the Lower cards. In the current Mudfish design, the DWL coincides with the chine point by construction — this is a deliberate design choice, not a mathematical requirement. A second chine below DWL would simply add another card to the Lower stack.

§4 The Golden Layout

The canonical spreadsheet layout is `four_quadrant_mantra.xlsx`, established September 15, 2026. All future additions must fit into this layout — not replace it.

Layout structure:

- Forward ($X > 0$) and Aft ($X < 0$) quadrants sit side by side horizontally.
- Upper ($Z > \text{DWL}$) and Lower ($Z < \text{DWL}$) halves stack vertically, separated by a visible DWL divider row.
- Each quadrant contains an independent column of matrix cards reading top-down: Sheer $\rightarrow$ DWL $\rightarrow$ Chine $\rightarrow$ Keel.
- The DWL card is simultaneously the bottom card of the Upper stack and the top reference of the Lower stack.

Governing rule: incremental refinement only. Never redesign the layout from scratch.

§5 Card Architecture

5.1 Two-Layer Separation

Every card has two independent layers that never mix:

Formula Layer (left side): Quadratic interpolation formulas, hull parameters, named cells. Produces one clean homogeneous point $(X, Y, Z, 1)$.

Operator Layer (right side): 4×4 homogeneous matrix M . Consumes the point from the formula layer. Produces $P' = P \times M$.

These layers communicate through exactly one handshake: a clean $(X, Y, Z, 1)$ row vector. Adding a new formula never touches a matrix. Adding a new operator never touches a formula.

5.2 Card Types

ANCHOR: Sets an absolute position directly. Input P is the actual global point. Matrix $M = \text{Identity}$ (no transformation). Used for the first card in each quadrant stack (currently: SHEER).

CHAIN: Rotates a local direction vector and translates it to the previous card's endpoint. Input P is local $(0, \cos \cdot L, \sin \cdot L, 1)$. Translation row of $M =$ previous card's output point.

5.3 Card Stack (current Mudfish design)

- SHEER (ANCHOR) — highest point, $+Z$ top
- DWL (CHAIN) — waterline, $Z=0$ in local frame
- CHINE (CHAIN) — intersection of topsides and underbody
- KEEL (CHAIN) — centerline, $Y=0$ by construction

5.4 Modularity Rule

To add a chine: copy one CHAIN block, paste it between the existing chine and keel cards, set its angle and length laws. Nothing else changes. The chain resolves by CardID lookup (INDEX/MATCH), not by row position, so copy-paste cannot silently re-wire the stack.

§6 Matrix Kernel — Verified Invariants

Status: VERIFIED September 14, 2026 by KernelTests sheet + independent hand calculation + independent ChatGPT inspection.

Five kernel invariants, all currently PASS in the live workbook:

TEST-001: Identity: $[1,0,0,1] \times I = [1,0,0,1]$

TEST-002: +90° Z rotation: $[1,0,0,1] \times \text{Rz}(90^\circ) = [0,1,0,1]$ (confirms +Y=port, right-hand rule)

TEST-003: Translation: $[1,2,3,1] \times T(10,20,30) = [11,22,33,1]$

TEST-004: Port/Stbd mirror: $[5,7,3,1] \times \text{My} = [5,-7,3,1]$

TEST-005: Composition order: $P \times \text{Rz} \times T \neq P \times T \times \text{Rz}$ (order matters, confirmed noncommutative)

Tolerance policy: $\text{eps_C0} = \text{eps_C1} = 0.001\text{mm}$ (explicit named cells in KernelTests sheet, not hardcoded literals).

§7 Hull Continuity — Verified

Status: VERIFIED September 14-15, 2026 by HullContinuityFixture v2.

The four-quadrant surface patches use a shared-skeleton algebraic family that guarantees exact boundary continuity by construction:

$$F(X, Z) = B0 + p \cdot X + q \cdot \zeta + A \cdot X^2 + B \cdot \zeta^2 + C \cdot X^2 \cdot \zeta^2$$

where $\zeta = Z - Z_{\text{DWL}}$

Shared terms (p, B0) force exact C0 and C1 continuity at midship and DWL boundaries. Independent terms (A, C) allow each quadrant to differ away from the boundaries.

Five conditions verified by live PASS/FAIL formulas in HullContinuityFixture:

- Midship C0+C1: F-U meets A-U, F-L meets A-L — smooth joins, PASS
- DWL C0: position continuity across the waterline — PASS
- DWL C1: intentional chine slope break — correctly labeled as expected discontinuity
- Negative Control 1: pure position gap (delta offset) — DETECTED
- Negative Control 2: pure tangent kink ($\gamma \cdot X$ perturbation, C0 preserved) — DETECTED

Key principle: three sampled PASS rows prove global continuity only when the algebra guarantees it everywhere. When real spline patches replace the skeleton family, dense sampling must be added as backstop.

§8 Sheer Formula — Specified, Not Yet Built (September 15, 2026)

The sheer line is the intersection of the XY plan view (half-breadth taper) and the XZ profile view (sheer rise curve). Both are quadratic formulas in X, same family as the bow taper and keel rocker.

8.1 Named Parameters

FB_Min: Minimum freeboard height (mm) — actual lowest point of the sheer

X_MinFB: Longitudinal position of FB_Min (mm). 0 = midship. Positive = shifted toward bow. Negative = shifted toward stern.

FB_Bow: Freeboard at bow tip (mm) — highest point

FB_Stern: Freeboard at stern (mm) — typically between FB_Min and FB_Bow

8.2 Derived Coefficients

$$\begin{aligned}C_{\text{sheer_fwd}} &= (\text{FB_Bow} - \text{FB_Min}) / (\text{BowX} - X_{\text{MinFB}})^2 \\C_{\text{sheer_aft}} &= (\text{FB_Stern} - \text{FB_Min}) / (\text{SternX} - X_{\text{MinFB}})^2\end{aligned}$$

8.3 Sheer Z Formula

$$\begin{aligned}Z_{\text{sheer_fwd}}(X) &= \text{FB_Min} + C_{\text{sheer_fwd}} \times (X - X_{\text{MinFB}})^2 & [X \geq X_{\text{MinFB}}] \\Z_{\text{sheer_aft}}(X) &= \text{FB_Min} + C_{\text{sheer_aft}} \times (X - X_{\text{MinFB}})^2 & [X \leq X_{\text{MinFB}}]\end{aligned}$$

Setting $X_{\text{MinFB}} = 0$ recovers the symmetric midship-minimum case. Setting $\text{FB_Min} = \text{FB_Bow} = \text{FB_Stern}$ recovers a flat deck. The formula handles all cases with no branching, no IF statements.

8.4 Reference Length

The rise percentage is referenced to LWL (half), not LOA and not FreeBoardMax. Rationale: LWL governs the hydrodynamic environment; LOA includes overhangs that extend beyond the waterplane and would artificially flatten the sheer curve on long-overhang designs.

$$\begin{aligned}\text{SheerRise_Bow} &= \text{BowRisePct} \times \text{BowX} & (\text{BowX} = \text{forward half of LWL}) \\ \text{SheerRise_Stern} &= \text{SternRisePct} \times \text{SternX} & (\text{SternX} = \text{aft half of LWL})\end{aligned}$$

8.5 Integration with Card Layout

The sheer formula lives entirely in the Formula Layer (left side of card). The Operator Layer (matrix side) is unaffected. The formula produces one clean (X, Y, Z, 1) point which feeds into the matrix operators exactly as today.

Status: EARMARKED. To be wired into the golden layout in a future session.

§9 View Operators vs Section Operators — Distinction Locked

Two fundamentally different operations must never be conflated:

View Operator (projection): Selects components from an already-transformed point for orthographic display. XY view = (X,Y). XZ view = (X,Z). YZ view = (Y,Z). Implemented as simple coordinate selection — no matrix multiply needed for pure orthogonal views.

Section/Intersection Operator: Finds where the hull surface actually intersects an oblique plane. Requires root-solving or curve extraction, not just coordinate selection. A projection matrix alone does not find an intersection curve.

Oblique views (diagonal plane cuts, buttock planes at arbitrary angles) require a real matrix multiply and may require iteration. These are future work.

§10 Open Items and Next Steps

As of September 15, 2026:

- Wire the sheer formula (§8) into the golden layout as live formula cards feeding the matrix operators.

- Add sheer rise visualization: XZ profile view and XY plan view reading left-to-right in the matrix-visualization style ($P \times M_1 \times M_2 = P'$).
 - Ballast and stability calculation — flagged as first priority from project inception, still unaddressed.
 - Confirm Hull Continuity Fixture sampling strategy once real hull equations replace the skeleton family placeholders.
 - Keel rocker formula — same quadratic family as sheer, independent rise percentage fore and aft, LWL-referenced. Specify and wire in.
 - Five confirmed FreeCAD bugs from earlier sessions — consider contributing fixes as open-source contributions under Mudfish's engineering-log philosophy.
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Amendment Log

September 15, 2026 — Initial specification created. All sections §1-§9 established.

(Future amendments appended below with date and section reference.)